

Hamza Azeem

TECH LEAD | SOFTWARE ENGINEER

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PROFESSIONAL EXPERIENCE

Amazon | SWE, Ads Technical Lead

Ads Reach Engineering

Dec. 2020 – Present

- Founding engineer and technical lead for Amazon Ads Reach, setting architecture and measurement methodology across campaign forecasting, historical analytics, identity, and data quality; the platform underpins DSP forecasting, media planning, executive reporting, and external agency integrations
- Replaced the legacy one-dimensional reach estimator with a multi-period supply-matrix and probabilistic sampling architecture, decreased forecast error 50%+ across core products and cutting it by 80% for programmatic guaranteed campaigns
- Re-architected global reach forecasting into an independently deployable platform and gave each engineer an isolated production-parity stack, cutting time to production from 9-15 days to about 1 hour and local iteration from 7+ minutes to seconds
- Established Amazon Ads' source-of-truth historical reach platform across 8 marketplaces and 90+ targeting segments, powering measurement for Prime Video, Twitch, Streaming TV, Audio, Devices, and online video; reduced segment onboarding from 1 week to half a day and compute cost per run by 99%
- Established standardized quality gates and automated validation for 2 PB+ daily across 20+ pipelines, reducing release approval from days to hours for executive and externally published reach data
- Built a sub-200 ms real-time reach and frequency API over a multi-TB Lucene index, now a core Amazon DSP forecasting dependency; launched an Open Data API adopted by major global advertising holding companies

Search Data & ML Platform

Aug. 2018 – Nov. 2020

- Created Amazon's first low-latency, hourly attributed shopping-behavior dataset, enabling fine-grained model training and analysis for millions of customers
- Built a centralized marketplace data and onboarding platform, reducing resource cost by 79% and improving turnaround 5x across Search datasets
- Led global GDPR automation for Amazon Search, eliminating 20+ hrs. per week manual work for thousands of data-subject requests
- Reduced schema-change costs on a petabyte-scale master dataset by 90% through Avro adoption, MapReduce optimization, and migration to AWS

Visual Search

2016 – 2017

- Developed and deployed the Visual Search team's first machine-learned barcode relevance-ranking models, increasing match rate by 14% and exceeding S-Team target
- Designed and launched a near real-time ingestion pipeline and API that reduced catalog-update latency from 3 days to under 5 seconds
- Built a backend service supporting near real-time additions, corrections, and deletions in Amazon's barcode-search catalog

CORE EXPERTISE

- Distributed Systems
- Forecasting & ML Infrastructure
- Data Platforms
- Real-Time API architecture
- Cross-org technical direction

TECHNOLOGIES

- Java, Python, SQL, Scala, R
- **AWS**
EMR, Glue, Step Fns
EC2, IAM, Lambda,
SageMaker Model
Serving,
CloudFormation
- Spark
- Lucene

EDUCATION

University of Waterloo

B.A.Sc. in Computer
Engineering
2013 – 2018